GUJARAT TECHNOLOGICAL UNIVERSITY

SMT. S. R. PATEL ENGINEERING COLLEGE, DABHI.

A Report On-

"Library Management System"

Under subject of

DESIGN ENGINEERING

B.E. Semester (V)

(Computer Engineering)

Submitted by: -

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Academic year (2025-26)

SMT. S R PATEL ENGINEERING COLLEGE, DABHI-UNJHA
GUJARAT TECHNOLOGICAL UNIVERSITY, AHMEDABAD

DEPARTMENT OF COMPUTER ENGINEERING

Certificate

This is to certify that we have examined the project titled.

"Library Management System"

Being submitted by

1.Panchal Kashish (240783107003)
2.Suthar Jahnvi (240783107009)

Is Undergraduate Students of Computer Engineering. I hereby accord our approval of it asa study carried out and presented in manner required for its acceptance in Partial fulfilmentof the project assigned to Semester (5th) for which it has been submitted. This approval doesnot necessarily endorse or accept every statement made, opinion expressed, or conclusion drawn as recorded in this project. It only signifies the acceptance of the project for the purpose for which it is submitted.

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ACKNOWLEDGEMENT

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Thanks.

Mr. Amit Prakash Tiwari
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ABSTRACT

Library Management System is an android application for everyone to buy and sell books through online.

Library Management System targeted the college students mainly. The main objective is that users can buy, sell & rent a book. Before this application implementation the manual process is used to do the processes of buy, sell, and search the books to everyone like students, faculties, writers etc. This manual process requires man power and more time consuming. To avoid such difficulties, we implemented this system.

When user's login to the system they can search for a book by their name, price, and Authors. Users also return a book and if they are late to return book, they need to give penalty. Admin have all the permission to access the record.

❖ Introduction:

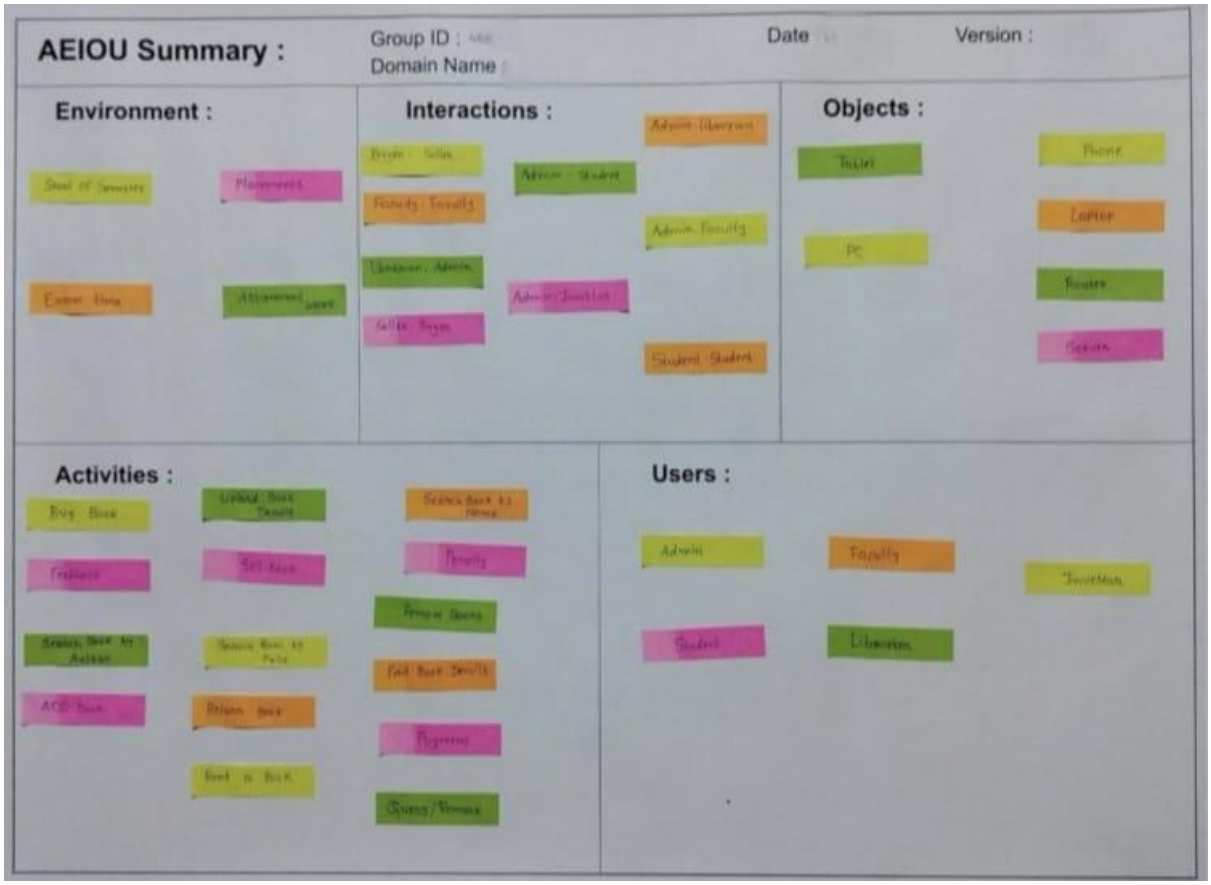
- This Library Management System is developed in favor of the Library Management System team which helps them to save the records of the students about their rooms and other things. It helps them from the manual work from which it is very difficult to find the record of the students and the information about those ones who had left the hostel years before.

- This solution is developed on the plight of the Library Management System team, through this they cannot require so efficient person to handle and manage the affairs of the students in the hostel, all you need to do is to login as administrator and you can see the information of all the students who have obtained and registered their hostel form, click verify to ascertain their eligibility and allocate them to the available hostel.

- Identification of the problems of the existing hostel management leads to the development of computerized solution that will be compatible to the existing hostel management with a solution which is more users friendly and more GUI oriented. We can improve the efficiency of the hostel management, thus overcome the drawbacks of the existing management.

1. AEIOU Summary:

AEIOU stands for 5 elements to be coded: Activity, Environment, Interaction, Object, and User. Activities are goal-directed sets of actions—paths towards things people want to accomplish. ... Interactions are between a person and someone or something else; they are the building blocks of activities.



❖ Activity:

Activities are the various tasks which are done by the users and stakeholders. These affect the domain.

- Login
- Buy Books
- Upload Book Details
- Search Book by Name
- Search Book by Author
- Search Book by Price
- Penalty
- Book
- Remove Book
- Edit Book details.
- Rent Book
- Return Book
- Query/Remark
- Payment

❖ Environment:

It includes the entire arena where activities take place.

In Summary of ENVIRONMENT, we describe general impressions, elements, and features.

- Start of Semester
- Placements
- Exam Time
- Assignment Work

❖ Interaction:

These are between a person and someone or something else; they are the building block of activities.

In Summary of INTERACTIONS, we describe interactions that how people should interact with each other.

- Seller To Buyer
- Admin To Student
- Admin To Faculty
- Admin To Librarian
- Admin To Investor
- Student To Student

❖ **Objects**:

To maintain the flexibility, it is necessary to place required amount of object in required proportion in required position and for that there is requirement of a keen understanding of all the object placed, and that can be seen in the presented sheet. Here we describe general impression and elements:

- Smart Phone
- Laptop/PC
- Server
- Tablet

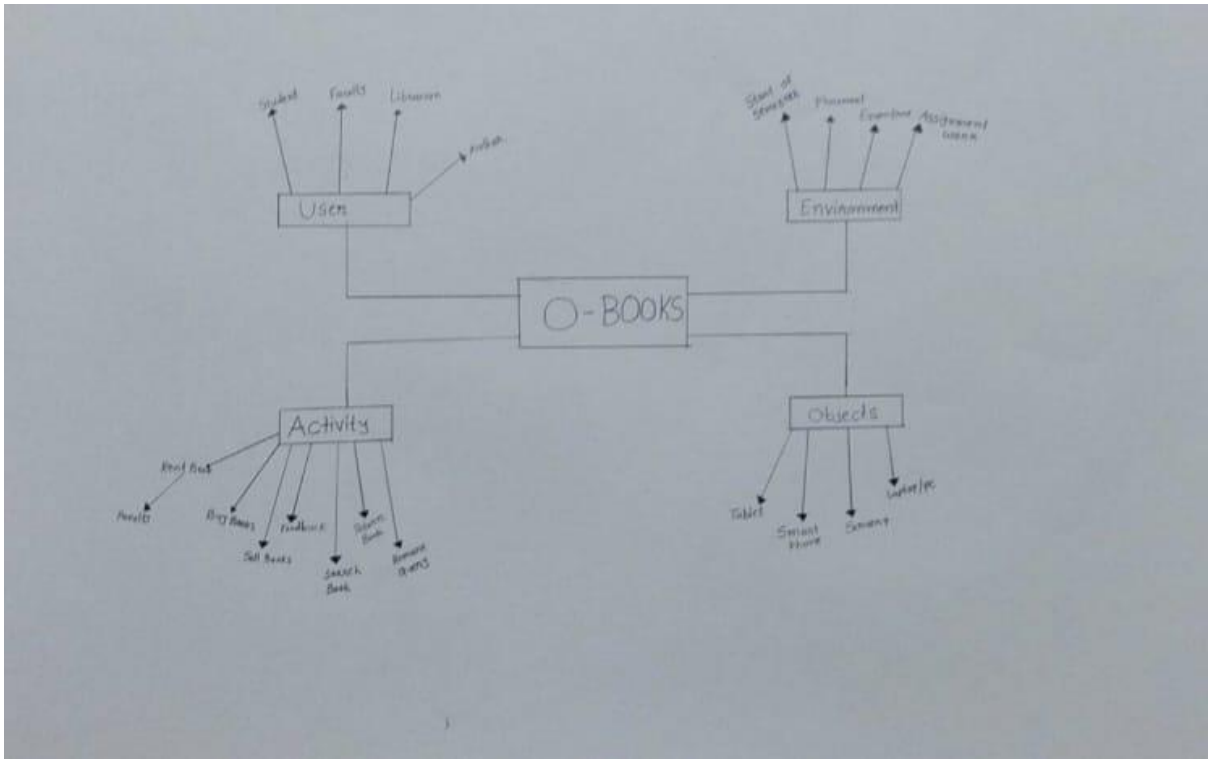
❖ **Users**:

Users are the people whose behavior's, preferences, and needs are being observed. In Summary of USERS, we describe in general impression and elements.
In general impression we describe this as the individual who participates in the process.

- Admin
- Student
- Faculty
- Librarian
- Investor

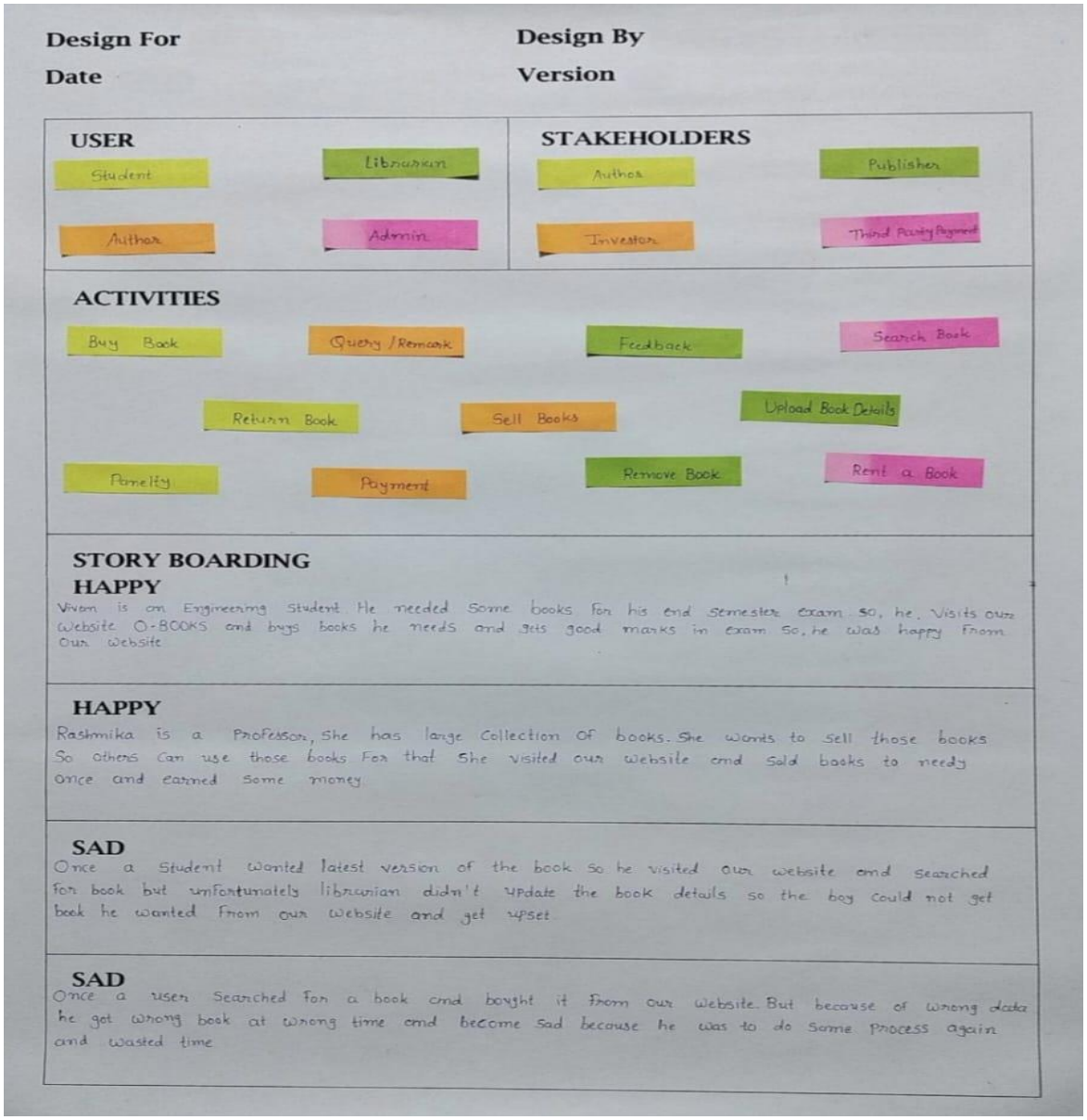
2. Mind Mapping Canvas:

A mind map is a graphical way to represent ideas and concepts. It is a visual thinking tool that helps structuring information, helping you to better analyses, comprehend, synthesize, recall, and generate new ideas.Just as in every great idea, its power lies in its simplicity.



3. Empathy Canvas:

From the empathy canvas we came to know which types of people are going to get benefits from our project. Which types of activities are done with the help of our project? Which problems werefaced by the users before our projects was designed? We came to know that were we can use our project. We got many ideas how to start the framework of any project.



❖ Users:

In an empathy canvas it shows you the details or the basic summary of multiple behavioral relation between different objects in this case users have different empathetic relations like.

- Students
- Admin
- Librarian
- Faculty
- Owner

❖ Stack Holders:

This section defines the different empathetic behaviors of stakeholders in that particular domain.

This could be an individual or group of individuals or even some corporation.

These are below mentioned stakeholders.

- Author
- Investor
- Publishers
- Third Party Payment

❖ Activities:

This section of the empathy canvas determines all the activities performed by all participants.

Activities are as listed below:

- Login
- Buy Book
- Upload Book Details
- Sell Book
- Search Book
- Rent Book
- Remove Book
- Feedback
- Return Book
- Edit Book Detail

❖ Story Boarding:

HAPPY: Vivan is an Engineering student. He needed some books for his end semester exam. So, he visits our website L.M.S. and buys books he needs and gets good marks in exam. So, he was happy from our website.

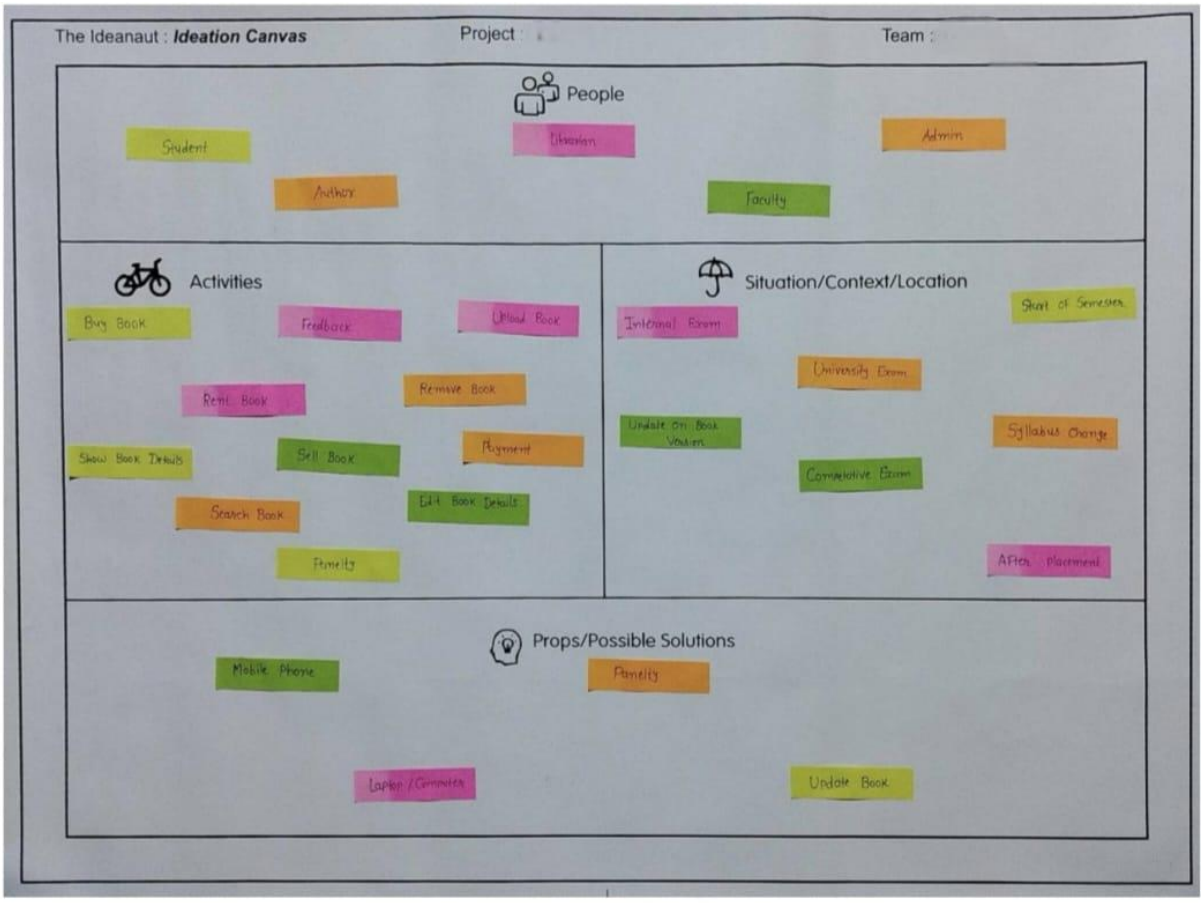
HAPPY: Rashmika is a professor, she has large collection of books. She wants to sell those books so others can use those books. For that she visited our website and sold books to needy once and earned some money.

SAD: Once a student wanted latest version of the book so he visited our website and searched for book but unfortunately librarian didn't update the book details so the boy could not get book he wanted from our website and get upset.

SAD: once a user searched for a book and bought it from our website. But because of wrong data he got wrong book at wrong time and become sad because he has to do same process again and wasted time.

4. Ideation Canvas:

Ideation Canvas is the creative process of generating, developing, and communicating new ideas, where anidea is understood as a basic element of thought that can be visual, concrete, or abstract. Ideation comprisesall stages of a thought cycle, from innovation, to development, to actualization.



❖ People:

- Student
- Author
- Librarian
- Faculty
- Admin

❖ Activities:

- Login
- Buy Book
- Upload Book Details
- Sell Book
- Search Book
- Remove Book
- Feedback
- Return Book
- Edit Book Details

❖ Situation/content/location:

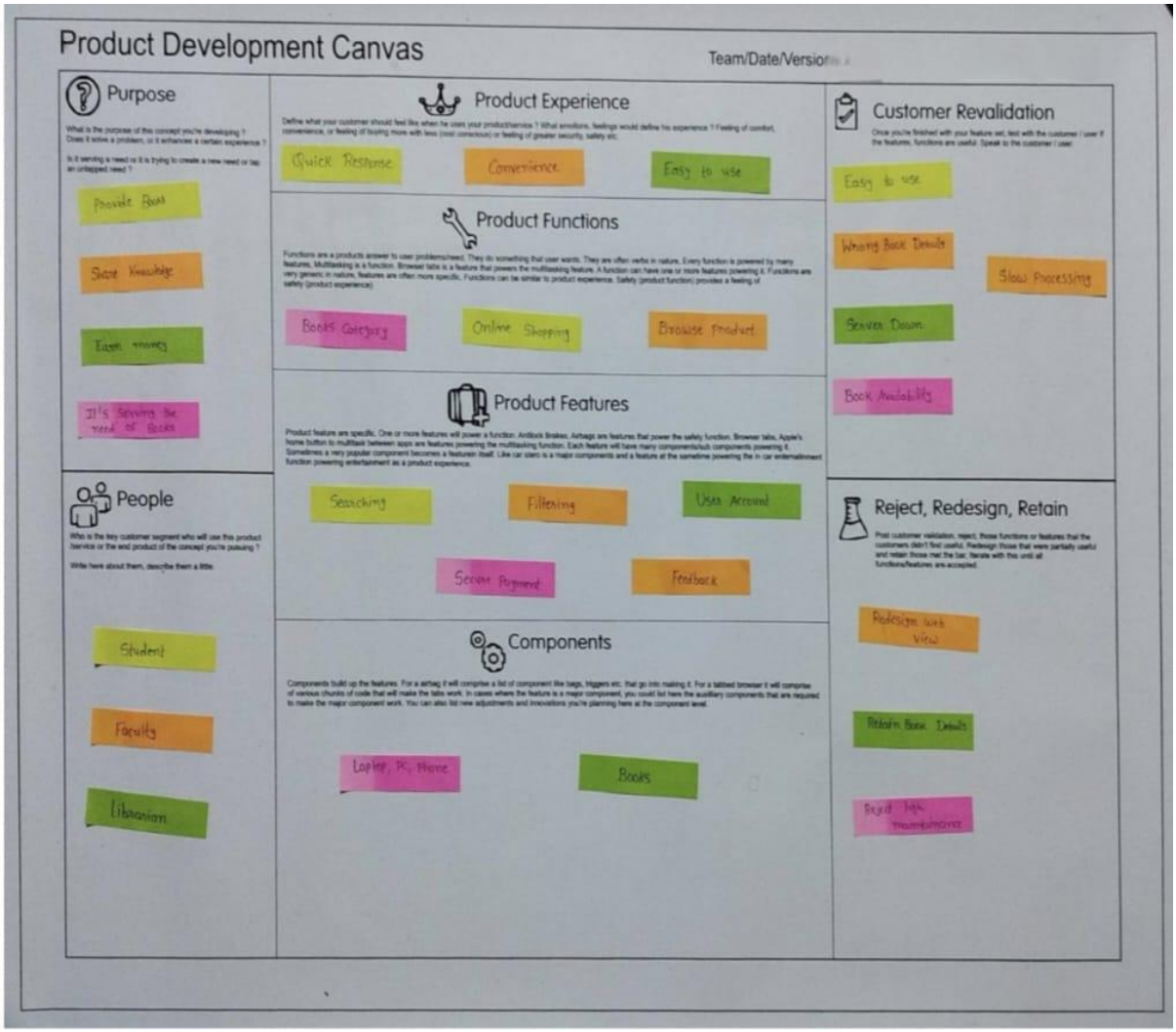
- University Exam
- Competitive Exam
- Start Of Semester
- Syllabus change
- After Placement
- Update on book version

❖ Props/Possible Solution:

- Mobile
- Laptop / PC
- Penalty
- Update book

5. Product Development Canvas:

Product development canvas can give you insights into these problems when creating a product, factors such as Purpose, people, product experience, product functions, product features, and components.



❖ Purpose:

- To provide Books
- To share knowledge
- To earn money

❖ People:

- Student
- Faculty
- Librarian
- Author

❖ Product Experience:

- Quick Response
- Easy to use.
- Convenience

❖ Product Function:

- Books Category
- Online Shopping
- Browse Product

❖ Product Features:

- Searching
- Filtering
- User Account
- Secure Payment
- Feedback

❖ Components

- Laptop/PC
- Mobile
- Data server
- Books

❖ Customer Revalidation:

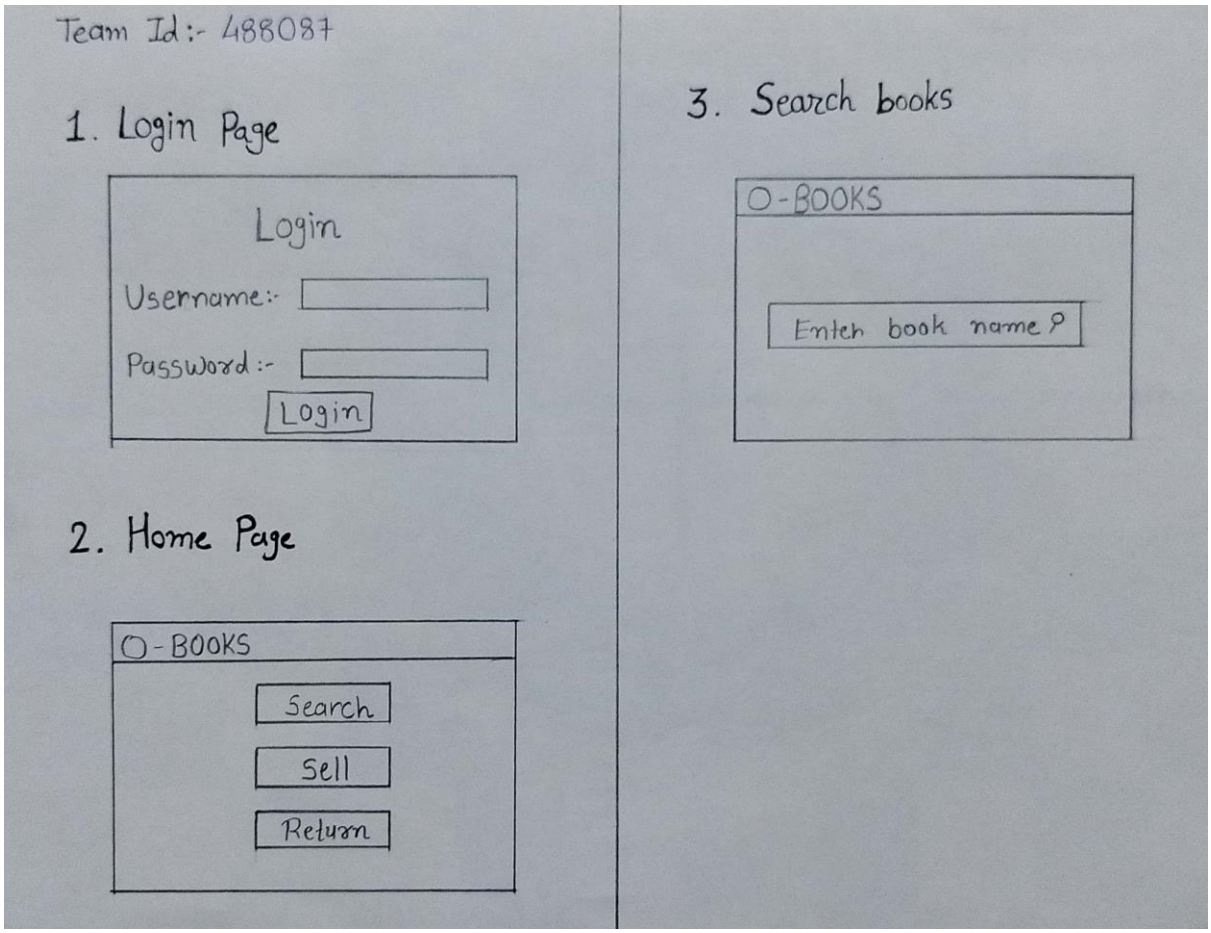
- Easy to use.
- Wrong Book Details
- Server Down
- Slow Processing
- Book availability

❖ Reject/Redesign/Retain:

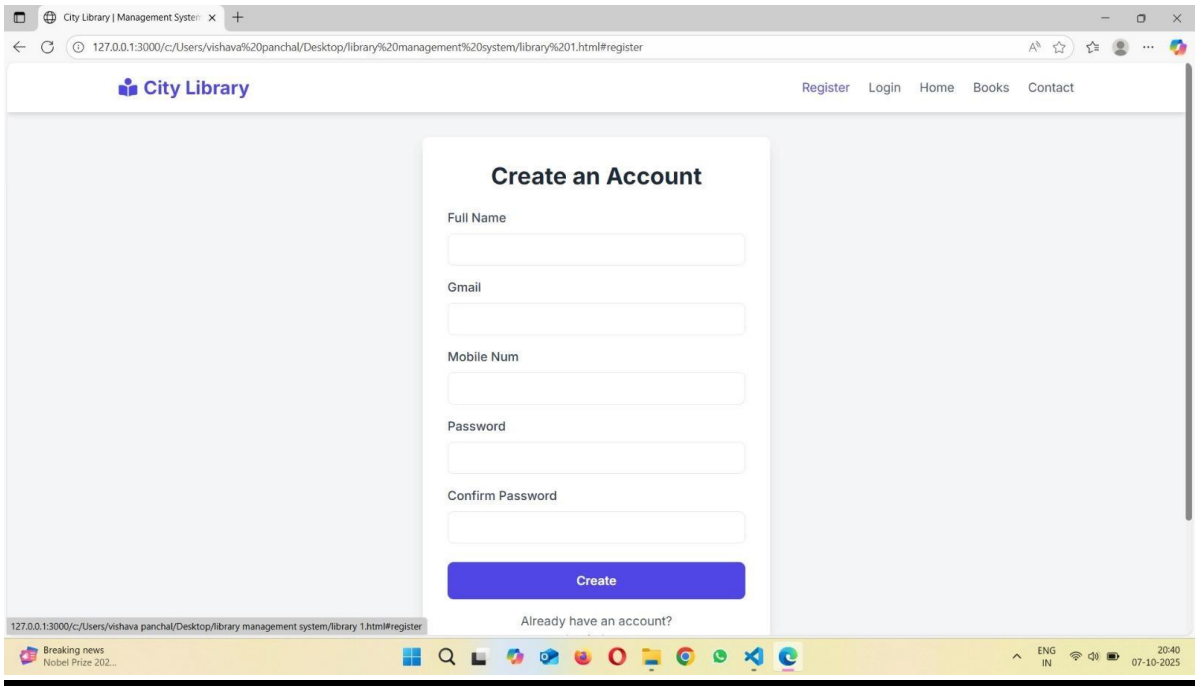
- Redesign web view
- Retain Book Details
- Reject High Maintenance

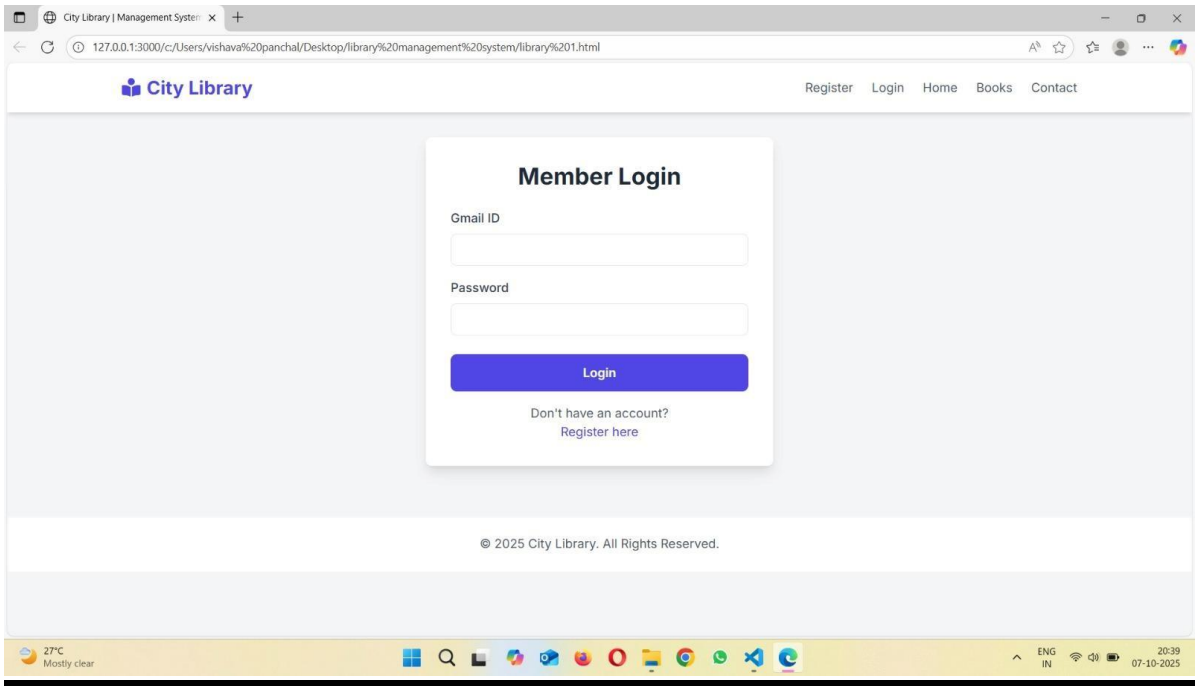
6. Prototype:

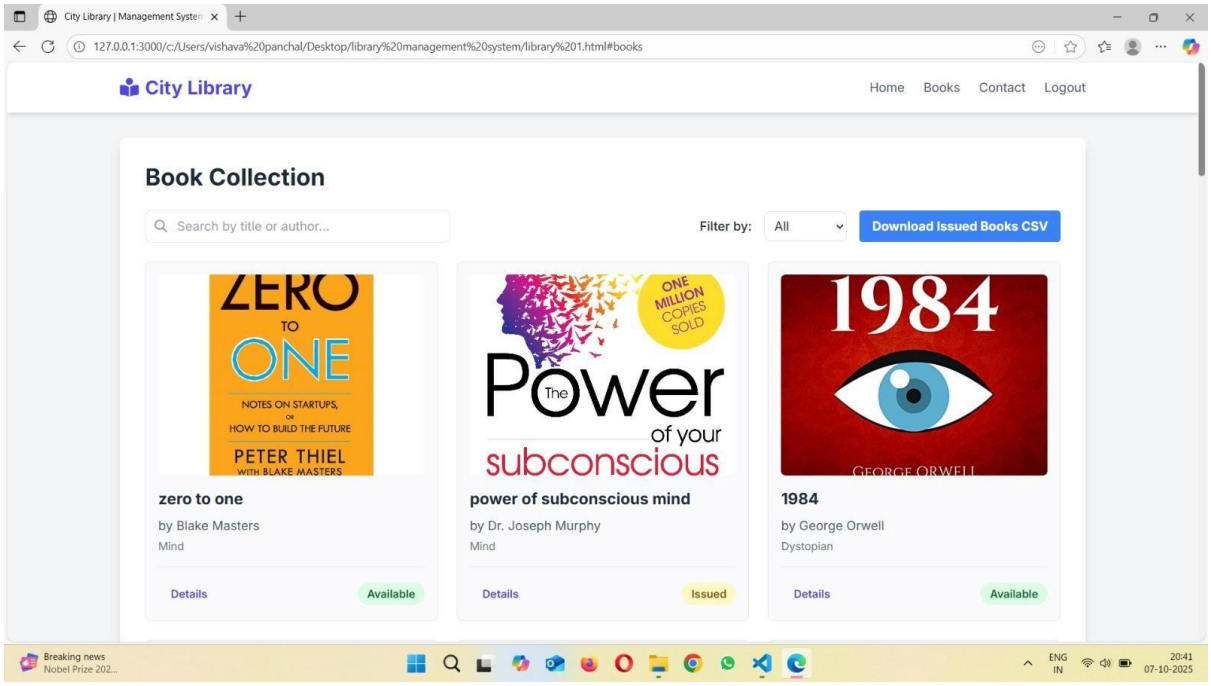
A prototype is an early sample, model, or release of a product built to test a concept or process.



WEBSITE MODULE



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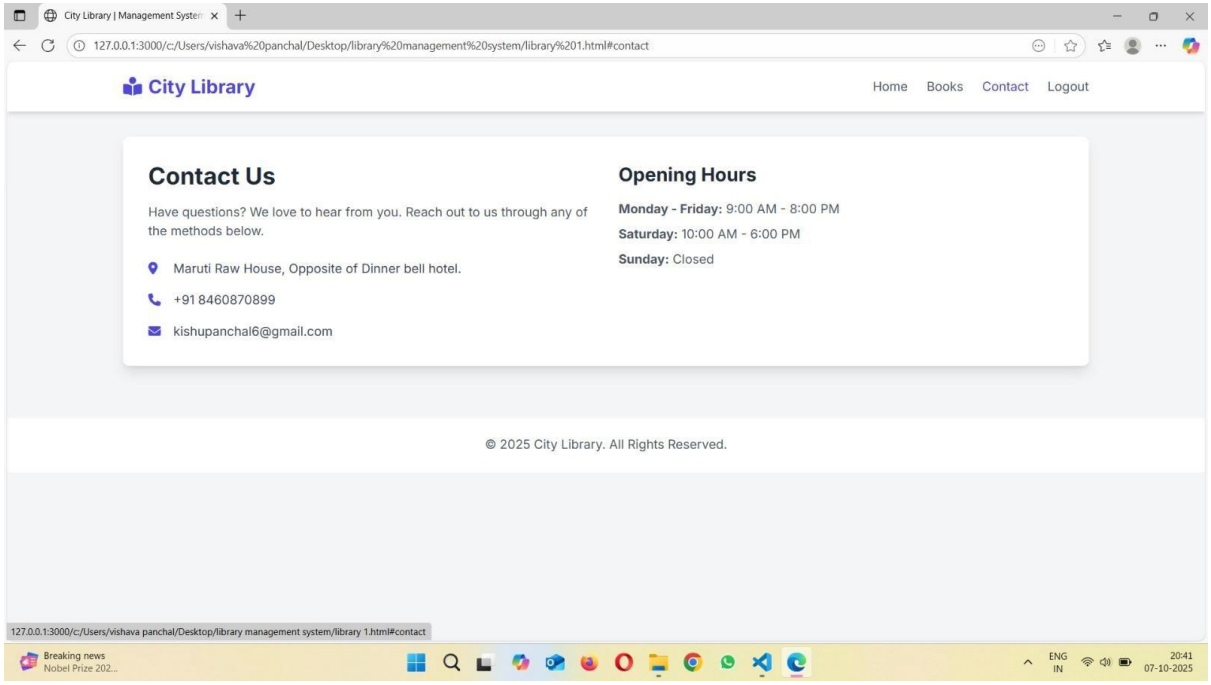
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Have questions? We love to hear from you. Reach out to us through any of the methods below.
Maruti Raw House, Opposite of Dinner bell hotel.
+91 8460870899
kishupanchal6@gmail.com
Opening Hours
Monday - Friday: 9:00 AM - 8:00 PM
Saturday: 10:00 AM - 6:00 PM
Sunday: Closed
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Conclusion

In the conclusion, we found the website that helps students to find their books in affordable price from other students, and as well as it also useful for those who have collection of books
and want to rent it.